Laser Crop Weeder Drone

Final Year Engineering Project

Project Title	Laser Crop Weeder Drone with Fixed Dock System
Department	[Your Department Name]
Guide / Mentor	[Faculty Guide Name]
Academic Year	[Year]

1. Team Overview

This project is developed by a team of 4 members. Each member owns a specific technical domain. All members share equally in field testing, report writing, and final presentation.

Sr.	Name	Role	Domain
1	Aditya	Team Head — Hardware Lead	Drone, dock, laser, all hardware
2	Christian	Hardware & Documentation	Dock support, wiring, full project report
3	Shaumik	AI Development Lead	Weed AI, crop health, edge AI on Raspberry Pi
4	Sanju	App Development Lead	SMS control, IVR, WhatsApp, Android app

2. Problem & Solution — Each Member's Contribution

Each member addresses specific real-world problems through their technical domain.

Member 1 — Aditya | Team Head & Hardware Lead

Problem he is solving:

- Farmers have no affordable, permanently stationed drone on their farm — they must carry, charge, and operate the drone manually every time.
- Drone nose modules are fixed — one drone can do only one job at a time.

- Drone landing after mission is imprecise — requires farmer to manually place it back.
- Wild animals destroy crops at night — no detection, no alarm, no deterrent.

His solution:

- Build a fixed solar dock on the farm — drone lives there permanently, always charged, always ready. Farmer never carries it anywhere.
- Design auto nose-swap mechanism in dock — drone changes its nose module (laser / spray / sensor / thermal) in one click without any manual work.
- Install IR LED beacon on dock for precision auto-landing — drone lands within 2–3 cm accuracy every single time.
- Add thermal camera night patrol mode — drone detects animals entering the farm after dark and triggers alarm with SMS alert to farmer.
- Assemble complete drone frame — motors, ESCs, Pixhawk flight controller, all sensors — and conduct all hardware safety tests.

Steps Aditya will follow:

1. Procure all drone components — frame, motors, ESCs, propellers, flight controller.
2. Assemble F450 drone frame and mount all motors, ESCs, and propellers.
3. Configure Pixhawk flight controller — calibrate IMU, GPS, compass, and radio.
4. Design and build fixed solar dock enclosure — frame, solar panel, battery bank.
5. Build auto nose-swap servo mechanism inside the dock.
6. Mount IR LED beacon on dock. Test precision landing with drone.
7. Build weatherproof auto-close dock lid with rain sensor.
8. Mount laser module, servo mirror, RGB camera, thermal camera, ultrasonic sensors.
9. Build mode dial and one-touch button on dock housing.
10. Conduct outdoor drone flight tests, hardware safety tests, and endurance tests.

Member 2 — Christian | Hardware Support & Documentation Lead

Problem he is solving:

- Hardware work is large — Aditya needs support for wiring, sensor integration, and farm field testing.

- Final year projects often fail at documentation — poor report and weak presentation lose marks even when the hardware works.
- Farmers need a simple user manual to operate the system — complex instructions make the system useless in a village.

His solution:

- Support Aditya on all dock electronics — battery health monitor, mode dial wiring, sensor connections, and pre-flight checklists before every field test.
- Operate Mission Planner GCS during all outdoor drone flights so Aditya can focus on hardware observation.
- Write the complete project report in proper college format — all 10+ sections from abstract to conclusion.
- Create presentation slides, architecture diagrams, block diagrams, and a clear user manual for the farmer.
- Maintain all project files — photos, code, test logs — in an organized shared folder accessible to all team members.

Steps Christian will follow:

11. Assist in dock wiring — battery monitor circuit, solar charging, mode dial circuitry.
12. Document every hardware build step with photos for the report.
13. Operate Mission Planner GCS during all outdoor drone test flights.
14. Write literature survey and introduction section of project report.
15. Draw system architecture diagram, block diagrams, and circuit schematics.
16. Write methodology, system design, and implementation sections.
17. Compile all test results, sensor accuracy data, and field logs into report tables.
18. Write conclusion and future scope section.
19. Create 20–25 slide presentation for college viva.
20. Write simple farmer user manual — how to press button, read SMS, and use the dock.

Member 3 — Shaumik | AI Development Lead

Problem he is solving:

- All existing drone weed-killing systems require GPS coordinates and pre-mapped field data before they can target weeds — not practical for Indian farmers.
- Farmers cannot identify crop disease or nutrient deficiency early — they see damage only after major yield loss has already happened.
- AI systems in drone projects require internet or cloud — they fail completely in villages with no connectivity.

His solution:

- Build vision-only weed detection — camera sees the weed in real time using YOLOv8 AI model, laser fires directly at it. No GPS, no pre-mapping needed.
- Build crop-safe laser guard — if AI confidence is below threshold, laser does NOT fire. This protects crop from accidental burning.
- Train crop health classifier — detects pest damage, nutrient deficiency (nitrogen, potassium, magnesium), and fungal disease from leaf images.
- Develop disease spread prediction — uses infected zone location and wind data to warn farmer which plot will be affected 2–3 days in advance.
- Optimize all AI models to TensorFlow Lite so they run on Raspberry Pi inside the dock — fully offline, no internet required.

Steps Shaumik will follow:

21. Collect and label weed vs crop image dataset — minimum 1000 images.
22. Train YOLOv8 weed detection model on labeled dataset.
23. Write real-time inference script — camera feed → model → weed XY coordinates.
24. Build crop-safe guard logic — confidence threshold before laser fires.
25. Implement auto laser power adjustment — thin vs thick weed.
26. Train crop health model — pest, nutrient deficiency, disease classes.
27. Develop disease spread prediction module using zone and wind data.
28. Implement acoustic crop stress detection from microphone signal.
29. Generate weed density heatmap from post-flight detection data.
30. Optimize all models to TFLite and test on Raspberry Pi for offline inference.

Member 4 — Sanju | App Development Lead

Problem he is solving:

- Most drone systems require a smartphone app and internet to operate — completely unusable for rural Indian farmers with basic 2G keypad phones.
- Farmers cannot get field reports after a mission — no feedback means they don't know if the drone did its job correctly.
- Farmers don't know when to sell their crop or which market gives the best price — they lose income from selling at the wrong time.

His solution:

- Build SMS command system — farmer texts SCAN or SPRAY from any basic phone. Dock receives the message and launches the correct drone mission.
- Build missed call trigger — 1 missed call = start daily scan, 2 missed calls = start spray mission. Zero data cost, zero internet.
- Build USSD code menu — farmer dials *123# from any phone. Menu appears: 1-Scan, 2-Spray, 3-Status. Works like mobile banking with no internet.
- Build IVR voice system — after mission, dock calls farmer's basic phone and automated voice reads results in their local language (Hindi / Marathi / Kannada).
- Build profit predictor — combines yield estimate + mandi market price + input cost and sends the expected income as SMS to the farmer.

Steps Sanju will follow:

31. Set up SIM800L GSM modem and connect to Raspberry Pi via UART serial.
32. Write SMS command parser — reads incoming SMS and maps to mission commands.
33. Implement missed call trigger — count incoming calls and launch missions.
34. Integrate USSD *123# gateway for dial-based mission menu.
35. Build IVR voice call system using gTTS — record Hindi, Marathi, Kannada audio.
36. Implement WhatsApp API integration for Android phone users.
37. Build SMS alert generator — creates local language mission result messages.
38. Integrate mandi price API — daily crop price comparison SMS to farmer.
39. Build profit predictor — yield + input cost + mandi price = income SMS.
40. Build basic Android farmer app — mission dashboard and field health summary.

3. Step-by-Step Project Plan — All Members Together

Step	Task	Who	Outcome
1	Procure all components — drone parts, dock materials, sensors, GSM modem	Aditya + Christian	All parts ready
2	Assemble drone frame — motors, ESC, propellers, flight controller	Aditya + Christian	Flying drone
3	Build fixed solar dock — frame, solar panel, battery bank, auto-lid	Aditya + Christian	Working solar dock
4	Auto nose-swap mechanism + IR beacon precision landing	Aditya	Nose-swap and auto-landing demo
5	Collect weed and crop image dataset and label it	Shaumik	1000+ annotated images
6	Train YOLOv8 weed detection model and test on images	Shaumik	Weed detection model
7	Integrate laser module — servo mirror + real-time vision targeting	Aditya + Shaumik	Laser kills weed on camera
8	Train crop health model — pest, disease, nutrient deficiency	Shaumik	Crop health classifier
9	Set up GSM modem — SMS command parser + missed call trigger	Sanju	Phone control working
10	Build USSD *123# menu and IVR voice call system	Sanju	USSD and voice call demo
11	Build WhatsApp integration and Android farmer app	Sanju	App working on phone

12	Optimize all AI models to TFLite — test offline on Raspberry Pi	Shaumik	Offline AI on Raspberry Pi
13	Full system integration — dock + drone + AI + phone control	All Members	Integrated system
14	Outdoor farm field test — weed kill, SMS control, night patrol	All Members	Field test report
15	Final report writing, presentation slides, and college submission	Christian + All	Submitted project

4. Work Distribution Summary

Lead = primary owner. Support = assists. — = not involved.

Task	Aditya	Christian	Shaumik	Sanju
Drone assembly & flight controller	Lead	Support	—	—
Solar dock construction	Lead	Support	—	—
Laser module hardware	Lead	—	—	—
Auto nose-swap mechanism	Lead	Support	—	—
IR beacon precision landing	Lead	—	—	—
Weatherproof dock lid	Lead	Support	—	—
Weed detection AI model	—	—	Lead	—
Laser vision targeting logic	Support	—	Lead	—
Crop health AI model	—	—	Lead	—
Edge AI setup on Raspberry Pi	Support	—	Lead	—

SMS & missed call control system	—	—	—	Lead
USSD & IVR voice system	—	—	—	Lead
Android farmer app	—	—	—	Lead
Profit predictor & mandi price API	—	—	—	Lead
Animal intrusion & night patrol	Lead	Support	—	—
Farm field testing	Lead	Support	Support	Support
Project report writing	—	Lead	—	—
Presentation slides & diagrams	—	Lead	—	—
User manual for farmer	—	Lead	—	—
Final viva and project demo	Lead	Support	Support	Support

5. Month-by-Month Timeline

Month	Aditya	Christian	Shaumik	Sanju
1–2	Procure parts, drone assembly	Literature review, help assembly	Collect & label dataset	GSM modem setup, SMS test
3	Build solar dock frame & circuit	Write intro & methodology sections	Train weed model V1	SMS command parser
4	Auto nose-swap + IR landing	Document hardware build with photos	Weed model V2 + crop-safe guard	Missed call trigger
5	Laser module + vision targeting	Draw diagrams & schematics	Crop health model training	USSD & IVR voice system
6	Full dock integration test	Compile test results for report	Disease spread + heatmap	WhatsApp + Android app

7	Farm field test (hardware)	Write results & discussion	Optimize models to TFLite on Pi	Profit predictor + mandi API
8	Bug fixing, weather test	Write conclusion & future scope	All AI farm accuracy testing	Full communication pipeline test
9–10	Final demo preparation	Final report + slides submission	AI documentation + accuracy report	App documentation + demo

6. Contribution Percentage

All 4 members contribute equally — 25% each. Domain work differs but total effort is equal.

Name	Domain Work	Shared Work	Total
Aditya	Drone, dock, laser hardware, sensors, field test lead	Report review, testing, presentation, demo	25%
Christian	Dock wiring support, pre-flight checks, full report, slides, manual	Testing, demo preparation, submission	25%
Shaumik	Weed AI, crop health AI, edge AI on Raspberry Pi, model optimization	Report AI section, accuracy tests, demo	25%
Sanju	SMS/call/USSD/IVR/ WhatsApp, Android app, mandi API, profit predictor	Communication tests, app demo, report	25%

7. Guide Approval

	Details
Guide Name & Designation	[Name] — [Designation]
Guide Signature	
HOD Signature	
Date of Approval	

Aditya Head — Hardware	Christian Hardware & Docs	Shaumik AI Development	Sanju App Development
[Signature]	[Signature]	[Signature]	[Signature]